ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PRACTICE EXAM 6 (PAPER 2) · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Basic Principles of Computer Programming and Computer Literacy practice exam 6 (paper 2).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Programs are not marked strictly on syntax: give marks for correct logic, variables and output. Accept other names, prompts and correct ways of solving the problem.

QUESTION 1

1.1 Project created and named ✓ (1)

1.2 Suitable sprite ✓ (1)

1.3 Suitable backdrop ✓ (1)

1.4 Variables speed, limit, over, fine created ✓✓✓✓ and set to 0 ✓✓✓✓ (8)

1.5 ask block ✓ with a clear question ✓; set speed to answer ✓ (3)

1.6 ask block ✓ with a clear question ✓; set limit to answer ✓ (3)

1.7 set over to (speed − limit) ✓✓ (2)

1.8 if over < 1 (not more than 0) ✓ then fine = 0 ✓ and say "No fine" ✓; else ✓ if over < 21 (not more than 20) ✓✓ then fine = 500 ✓; else ✓ fine = 1500 ✓; correct nesting ✓ (10)

1.9 say over ✓ and fine ✓ joined with text ✓ (3)

1.10 58 in 60: No fine ✓; 75 in 60: 15 over, R500 ✓; 145 in 120: 25 over, R1 500 ✓ (3)

```
when green flag clicked
set [speed] to 0
set [limit] to 0
set [over] to 0
set [fine] to 0
ask [What is the speed of the vehicle (km/h)?] and wait
set [speed] to (answer)
ask [What is the speed limit (km/h)?] and wait
set [limit] to (answer)
set [over] to ((speed) - (limit))
if < (over) < 1 > then
    set [fine] to 0
    say [No fine] for 2 seconds
else
    if < (over) < 21 > then
        set [fine] to 500
    else
        set [fine] to 1500
    say (join (join [Over the limit by ] (over)) (join [ km/h. Fine: R] (fine))) for 3 seconds
```

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QUESTION 2

2.1 File named correctly ✓ with a comment on the first line (1)

2.2 import turtle (and import random) ✓; pen = turtle.Turtle() ✓ (2)

2.3 pen.speed(0) ✓✓ (2)

2.4 pen.pensize(3) / pen.width(3) ✓✓ (2)

2.5 Outer for loop ✓ with range(36) ✓; correct indentation ✓; hexagon and turn inside the loop ✓✓ (5)

2.6 Inner for loop ✓ with range(6) ✓; forward(100) ✓; right(60) / left(60) ✓; inside the outer loop ✓ (5)

2.7 right(10) / left(10) after each hexagon ✓✓; random colour ✓✓ set before each hexagon ✓ (5)

```
# YourName - Question 2
import turtle
import random

pen = turtle.Turtle()
pen.speed(0)
pen.pensize(3)
turtle.colormode(255)

for i in range(36):                        # 36 hexagons
    pen.color(random.randint(0, 255), random.randint(0, 255), random.randint(0, 255))
    for side in range(6):                  # one hexagon
        pen.forward(100)
        pen.right(60)
    pen.right(10)                          # turn before the next hexagon

turtle.done()
```

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QUESTION 3

3.1 File created and named ✓ (1)

3.2 total_sales = 0 ✓✓ outside the function (2)

3.3 def sell_bundle(): ✓✓; name input ✓; size input as a number ✓; number of bundles as int ✓; global total_sales inside the function ✓ (6)

3.4 if size <= 5 ✓ price 30 ✓; elif size <= 20 ✓✓ price 25 ✓; else ✓ price 20 ✓; correct structure ✓ (8)

3.5 if bundles >= 3 ✓✓ cost multiplied by 0.9 ✓✓ (4)

3.6 cost = bundles × size × price ✓✓; receipt displayed ✓ with 2 decimals ✓ (4)

3.7 Loop (while) ✓✓; asks "Serve another customer?" ✓✓; stops when the answer is "no" ✓✓; correct indentation ✓ (7)

3.8 total_sales = total_sales + cost (or +=) ✓✓ inside the function using global ✓✓ (4)

3.9 sell_bundle() called in the loop ✓✓; total sales displayed after the loop ✓✓ (4)

3.10 Output R500.00, R324.00 and total R824.00 ✓✓ (2)

3.11 Saved and printed / presented ✓ (1)

```
# YourName - Question 3
total_sales = 0

def sell_bundle():
    global total_sales
    name = input("Enter customer name: ")
    size = int(input("Enter bundle size (GB): "))
    bundles = int(input("Enter number of bundles: "))

    if size <= 5:
        price = 30
    elif size <= 20:
        price = 25
    else:
        price = 20

    cost = bundles * size * price
    note = ""
    if bundles >= 3:
        cost = cost * 0.9
        note = " (10% discount)"

    print(f"Receipt for {name}: {bundles} x {size} GB at R{price} per GB = R{cost:.2f}{note}")
```

```
    total_sales = total_sales + cost

again = "yes"
while again == "yes":
    sell_bundle()
    again = input("Serve another customer? (yes/no): ").lower()

print(f"Total sales: R{total_sales:.2f}")
```

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TOTAL: 100

END OF MARKING GUIDELINE